

Emissions, Affordability, and Evidence under New York City’s Local Law 97

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Abstract

New York City’s Local Law 97 (LL97) places greenhouse-gas limits on large buildings in a housing system where retrofit costs, financing, timing, and benefits may be distributed unevenly. Evaluating this emissions–affordability problem is difficult because compliance, assistance, project delivery, and rent treatment are administered through different legal and institutional systems. This article uses a single-case documentary analysis to ask what the available evidence can establish about those connected processes. A frozen case corpus of 17 legal and administrative sources supports 37 material LL97 proposition clusters. Twenty-five concern formal design and 12 concern observable administrative activity. The analysis organizes them into six evidence stages: formal design; administrative activity; decision and timing; project delivery; verified compliance or emissions; and owner or resident outcomes. Three findings follow. Accommodation relevant to affordability is distributed across distinct institutions; legal route and housing cohort determine which records are comparable; and later claims about delivery, performance, or incidence require longitudinal linkage in the same building or population, route, and period. This paper presents an evidence-compatibility rule that separates the claim stage from the source authority and tests whether records can be joined before they support a combined emissions and affordability conclusion. Applied to the stricter 2030 period, the rule defines a cohort-sensitive design for future evaluation.

Keywords: building-performance standards; Local Law 97; energy justice; housing affordability; policy implementation; evaluability

1 Introduction

Buildings account for more than two thirds of New York City’s greenhouse-gas emissions. Local Law 97 (LL97) makes that sector a central site of climate policy: most buildings larger than 25,000 square feet became subject to greenhouse-gas limits beginning in 2024, with stricter limits taking effect in 2030 (New York City Department of Buildings, 2026f). The law therefore attaches the city’s decarbonization goals to particular properties, owners, reporting years, and forms of proof. In April 2026, the city reported filings from approximately 28,000 buildings under audit, placing the first compliance period in view while final audit results were still pending (New York City Department of Buildings and New York City Department of Housing Preservation and Development, 2026).

The same buildings belong to a heterogeneous housing system. A required project may create a capital and financing problem for an owner; its timing can affect procurement and compliance; and its consequences for residents may appear through rent treatment, energy bills, service reliability, comfort, health, or housing stability. These pathways do not automatically move together. Subsidized and market-rate buildings differ in physical systems and regulatory settings, while household energy burdens depend on linked building and socioeconomic conditions (Reina and Kontokosta, 2017; Pennell et al., 2022; Kontokosta et al., 2020a). Rental arrangements can also separate the party that pays for an efficiency investment from the party that receives some of its operating benefits (Singhal et al., 2025). The emissions–affordability problem is thus both substantive and institutional: it concerns how decarbonization is delivered and how its costs and benefits reach owners and residents.

LL97 brings this problem into sharp relief because the relevant decisions are distributed. The current Administrative Code contains an Article 320 regime of annual building emissions limits and reporting, along with an Article 321 regime of energy-conservation-measure requirements for a designated cohort (City of New York, 2026a,b). The Department of Buildings (DOB) administers the compliance and adjustment processes. The Department of Housing Preservation and Development (HPD), NYC Accelerator, and the Affordable Housing Reinvestment Fund (AHRF) describe the aid or funding routes. New York State Homes and Community Renewal administers a legally separate process for major-capital-improvement rent treatment. These institutions produce different records at different units and dates. A support offer shows that a route exists; a filing shows an administrative event; a completed project, verified emissions result, or rent decision answers a different question.

That difference creates an evaluability problem. Research on New York’s earlier benchmarking and audit policies estimates effects with defined populations, measured outcomes, and temporal or comparison designs (Meng et al., 2017; Kontokosta et al., 2020b; Papadopoulos

los et al., 2018). Energy-justice research likewise requires indicators that correspond to a specified distributive, procedural, or recognition claim (Jenkins et al., 2016; Baker et al., 2023). The central risk is an invalid join: combining records that refer to different legal routes, housing cohorts, periods, or actors, and treating the result as evidence of combined climate and affordability performance.

This article asks: What can a documentary analysis establish about the emissions–affordability problem under LL97, and what evidence would be required to evaluate implementation as the 2030 limits take effect? It argues that LL97’s substantive affordability challenge spans multiple rules and agencies. Accommodation is distributed across compliance, assistance, finance, project delivery, and rent treatment. The legal route and housing cohort determine which administrative records are comparable, and claims about delivery or incidence require those records to be linked longitudinally at the same building or defined population and period.

The article develops this argument through six evidence stages: formal design; observable administrative activity; decision and timing; project delivery; verified compliance or emissions; and owner or resident outcomes. The stages classify the proposition being assessed. Source authority, provenance, route and temporal fit, and linkage sufficiency are evaluated separately. This separation is the framework’s analytical contribution: it turns a familiar distinction between activities and outcomes into a compatibility test for a legally fragmented building mandate.

The analysis uses a transparent documentary single-case method. Section 2 connects building-policy evaluation, housing and energy justice, document analysis, and evaluability assessment. Section 3 defines the corpus and coding rules. Sections 4 and 5 develop three LL97 findings and a worked institutional trace. Section 6 translates them into a cohort-sensitive 2030 evaluation design. The conclusion explains what this case contributes to the interdisciplinary evaluation of environmental standards operating across fragmented institutions.

2 Literature and Analytical Framework

2.1 From building information to policy outcomes

Building-performance standards (BPS) sit within a wider policy mix of benchmarking, disclosure, audits, technical assistance, financing, performance targets, and enforcement. Each instrument creates a different object of analysis. Benchmarking makes specified building information reportable; an audit records diagnostic findings; a standard attaches an obligation

and date to a performance metric; and a completed retrofit changes a physical system. The information functions of benchmarking and disclosure are therefore distinct from their effects on decisions or performance (Palmer and Walls, 2017). Open building datasets also depend on coverage, field definitions, and documentation (Roth et al., 2020). Standards and disclosure can operate as complementary components of a policy mix, while target timing and scope shape modeled pathways (Berry et al., 2022; Walter and Mathew, 2021).

Outcome research shows what is needed to move from an administrative record to an effect claim. Studies of New York’s benchmarking policy, mandatory audits, and building-performance trajectories use defined populations, longitudinal observations, measured outcomes, and comparison or timing strategies (Meng et al., 2017; Kontokosta et al., 2020b; Papadopoulos et al., 2018). Different questions require different designs. A legal analysis can establish an obligation; an implementation study can establish what administrators or regulated actors did; a performance study requires a technical outcome aligned with the relevant building and period.

This alignment becomes harder when records cross systems. Energy certificates, reported benchmarks, completed audits, modeled trajectories, and verified emissions results differ in unit, period, and decision relevance even when they are grouped under the label “energy performance” (Lartigue et al., 2022). Building-data research describes a related “granularity gap”: systems designed for reporting or billing may lack the context needed to evaluate implementation (Syse et al., 2026). For an affordability claim, the required context also includes housing status, responsible actor, financing, and the mechanism through which a cost or benefit could reach an owner or resident.

2.2 Housing affordability and energy justice

The affordability side of a BPS requires multiple measures. In New York multifamily housing, energy use and system conditions vary according to building, socioeconomic, and regulatory characteristics (Reina and Kontokosta, 2017; Pennell et al., 2022). Kontokosta, Reina, and Bonczak combine building energy and audit records with small-area socioeconomic and household estimates to study energy-cost burdens (Kontokosta et al., 2020a); cross-domain housing research likewise depends on records that connect policies, properties, and communities (Asensio et al., 2024). For the present framework, the literature motivates separate indicators for capital, finance, project scope, rent treatment, energy expenditures, property responses, and resident conditions because the responsible actors and timelines differ.

Retrofit and rental-market scholarship identifies several mechanisms that matter for LL97

without determining their New York effects in advance. Household vulnerability can shape the experience of retrofit policy; subsidy uptake can be uneven; and rental arrangements can create split incentives (Gillard et al., 2017; Willand et al., 2020; Singhal et al., 2025). Evaluations of rental energy standards require linked property and market data, while analyses of possible BPS penalties remain conditional on local parameters and modeled assumptions (Fuerst et al., 2026; Taylor et al., 2025). These studies direct attention to project costs and finance, owner capacity, rent and energy expenditures, technical delivery, and resident experience.

Energy justice adds distributive, procedural, and recognition questions: how costs and benefits are allocated, how decisions are made, and whose circumstances and knowledge count (Jenkins et al., 2016). Decarbonization’s political and distributional conditions remain empirical even when its climate objective is clear (Sovacool et al., 2019), and a review of energy-justice methods finds substantial variation across places, technologies, and research designs (Jenkins et al., 2021). Indicator choice is part of that variation because a measure must correspond to the equity dimension and decision context it is intended to represent (Baker et al., 2023).

Resident-centered studies illustrate the difference. Charles and colleagues interview social-housing residents about retrofit experience, while Furman and Hadjri interview retrofit stakeholders about the inclusion of resident expertise (Charles et al., 2025; Furman and Hadjri, 2025). Mixed-method work on Boston’s building-performance standard combines documents, observation, and interviews to study implementation and climate justice directly (Diezmártinez et al., 2024). A filing count answers a different question by recording an administrative event.

2.3 Documentary evaluability and six evidence stages

Document analysis examines a document’s purpose, provenance, context, and limits while treating the event it describes as a separately evidenced object (Bowen, 2009). Evaluability assessment asks a related question before an impact study: whether a program’s design, implementation record, and available measures make the proposed evaluation useful and feasible (Trevisan, 2007). The present framework combines those traditions with building-policy evaluation and energy-justice measurement. Its purpose is to identify what kind of claim each LL97 record can support and whether records from different institutions can be joined for a combined emissions and affordability analysis.

Table 1 defines six evidence stages and their LL97 application. The stages classify the subject of a proposition; source quality is assessed separately. A current statute is

authoritative for formal legal design, whereas project delivery requires project records. A well-verified resident interview may directly document lived experience, whereas statutory content requires the controlling text. Source authority and provenance are therefore assessed separately from stage.

Table 1: Evidence stages and the LL97 questions they can answer

Evidence stage	Illustrative LL97 record	Question the record can answer
1. Formal design	Code provision, adopted rule, adjustment authority, or published support and finance route	What obligation, accommodation, eligibility rule, or institutional responsibility exists?
2. Administrative activity	Dated filing, audit status, Covered Buildings List entry, registry entry, or recorded program contact	What administrative event was recorded, for which population and date?
3. Decision and timing	Adjustment disposition, mediated agreement, approval or denial, and decision dates	What decision occurred, on what terms, and after what interval?
4. Project delivery	Procurement, contract, installation, commissioning, payment, or completion record	What physical work or financed intervention was delivered?
5. Verified compliance or emissions	Finalized Article 320 emissions record or route-specific Article 321 compliance proof	What verified technical or compliance result applies to the building and period?
6. Owner or resident outcome	Linked cost, financing, rent, energy-bill, service, health, participation, or housing measure	How did a defined owner or resident condition change?

The framework applies four compatibility tests to any proposed connection among stages. First, the source must have the authority and provenance needed for the claim. Second, the records must concern the same legal route and housing cohort. Third, their dates and reporting periods must align. Fourth, the unit of linkage must be a common building or defined population. A “formal accommodation” is a pathway, adjustment, support offer, or financing route at stage 1. Its availability is an important design fact; use, delivery, performance, and incidence are separate propositions.

The stages classify the object being observed without ranking source quality or imposing a fixed causal sequence. The framework’s added value lies in applying the four compatibility tests across institutions before an analyst combines those observations. This is more specific than a generic outputs–outcomes distinction: it preserves legal pathway, housing cohort, time, provenance, and linkage simultaneously in a fragmented environmental mandate.

3 Method

This study is a documentary single-case analysis of LL97’s evaluability. Its scope is legal architecture, recorded implementation activity, and the compatibility of evidence across institutions; causal effect estimation lies outside the design. The case was selected because a binding building-emissions mandate intersects with several housing cohorts, compliance timelines, assistance and finance arrangements, and a separate State rent-treatment system. This configuration makes the evidence problem visible within a substantively important climate policy.

The LL97 case corpus was frozen on August 2, 2026, and its reproducibility package was finalized on August 7. It contains 17 documents: current controlling law and adopted rules; official materials from DOB, HPD, NYC Accelerator, and New York State Homes and Community Renewal; AHRF program documents; and dated administrative sources including the Covered Buildings List (CBL), a property-data dictionary, registries, and agency releases. A separate set of 28 peer-reviewed articles supplies the conceptual, comparative, and methodological context; the case documents form the LL97 evidence base. A targeted currency check on August 22 confirmed that the cited Articles and rules remained current and located no later official release of finalized first-period audit results. The check left the frozen stage counts unchanged. It also identified an August 8 Department of Finance amendment permitting installment disbursements under the Property Assessed Clean Energy financing program, with interest accruing only on disbursed funds (New York City Department of Finance, 2026). The amendment changes finance administration while leaving the LL97 Articles and rules analyzed here unchanged; it is logged as a post-freeze source outside the coded corpus.

A case document was included when its issuer or author, date or version, jurisdiction, and permitted proposition could be identified. Current legal propositions follow an authority hierarchy: codified law, adopted rule, dated official guidance, other official program material, and then contextual scholarship. Guidance describes an agency’s published process; controlling law governs the legal proposition. Historical material is used only when its status is explicit. The case corpus excludes confidential material, anonymous accounts, unaudited recollections, and interviews.

The unit of analysis is a material LL97 documentary proposition. Closely related statements appear as one proposition cluster only when they share a source or source set, pinpoint, population, legal route, period, evidence stage, and inference boundary. The coding universe consists of every material LL97-specific proposition in the frozen case analysis. Contextual scale statements and the post-freeze currency check are logged separately and excluded from

the stage counts. The matrix is a claim audit of the selected corpus; its coverage is defined by these inclusion rules.

The six stages in Table 1 were specified in the study’s method plan and operationalized in a coding dictionary before final proposition coding. For each cluster, the matrix records the source and pinpoint, date or version, legal route, population or unit, institution, highest stage directly supported, permitted inference, and unresolved later-stage question. The highest-direct-support rule prevents promotion by sequence: a filing is administrative activity, a support offer is formal design, and a registry entry is the event recorded by that registry. A later stage requires a corresponding record.

The final matrix contains 37 proposition clusters. Twenty-five are supported at stage 1 and 12 at stage 2. Repeated uses of the same proposition are counted once, and the last three rows are case syntheses derived from the atomic entries. This distribution measures the evidentiary reach of the propositions in the frozen corpus. The prevalence of LL97 activities and the occurrence of later-stage events require separate data.

Legal, temporal, and linkage checks are applied in addition to stage coding. When sources differ, controlling text governs the legal proposition and the difference remains visible as an implementation observation. Each administrative quantity retains its stated denominator and date. Records are joined only when they refer to a compatible route, cohort, unit, and period. A failed search is recorded only as a dated, corpus-bounded nonlocation; the underlying event remains unresolved.

Online Resource 1 provides the source inventory, selection and exclusion rules, authority hierarchy, coding dictionary, 37-row source-to-claim matrix, bounded nonlocation and supersession log, and checksums for retained source copies. These materials make the selection and assignments auditable.

4 LL97’s Fragmented Affordability Architecture

LL97’s first case finding is that its affordability-relevant governance extends beyond the agency and rules that administer the emissions standard. Classification and compliance begin with City law and DOB, but technical assistance, finance, project delivery, and rent treatment sit in other institutional systems. The resulting evidence path is distributed across separate administrative files from obligation to owner or resident outcome. Recent LL97 scholarship places the law’s climate and public-health aims within a broader environmental-policy setting (Chervonski et al., 2025); the analysis here focuses on the institutional structure through which those aims are implemented.

4.1 Different routes create different forms of proof

The controlling legal distinction is between Article 320, which establishes annual greenhouse-gas emissions limits and reporting requirements for covered buildings, and Article 321, which establishes energy-conservation-measure requirements for a specified cohort (City of New York, 2026a, §§ 28-320.3, 28-320.7); (City of New York, 2026b, §§ 28-321.1–28-321.3.2). DOB’s public LL97 hub directs owners to the two Articles and their rules (New York City Department of Buildings, 2026f). These related parts of LL97 generate distinct evidence.

Article 320 organizes the principal emissions-limit and reporting pathway. Its implementing rule structures annual reporting and requires owners to retain the documents used to prepare a report for at least six years (New York City Department of Buildings, 2026b, § 103-14(b)). Under the Article 321 rule, an owner may demonstrate compliance either through an emissions report showing that 2024 emissions did not exceed the applicable 2030 Article 320 limit or through a certified report that the required prescriptive energy-conservation measures were fully implemented before January 1, 2025 (New York City Department of Buildings, 2026c, § 103-17(b)). Article 320 therefore generates recurring annual emissions reports; Article 321 permits a route-specific compliance demonstration. Counts from the two routes describe different proof and periods.

Classification can also change over time. DOB’s March 2026 CBL service notice distinguishes Article 320 cohorts beginning in 2024, 2026, and 2035 (CP0–CP2), Article 321 status beginning in 2024 (CP3), and City-owned or New York City Housing Authority status (CP4) (New York City Department of Buildings, 2026e, p. 1). A list entry reflects DOB’s records when published and may change after a dispute, partner-agency update, or change in building circumstances. The notice also gives transition treatment for specified 2026 path changes. A building moving to CP3 in 2026 under the stated conditions, for example, has a report due in 2027 and may seek a mediated resolution in 2027 if more time is needed (New York City Department of Buildings, 2026e, pp. 1–2). The adopted Article 321 rule remains keyed to May 1, 2025, for the report, eligible-project commitment, and ordinary mediated-resolution criteria (New York City Department of Buildings, 2026c, § 103-17(b), (f)–(g)). The CBL therefore records DOB’s published transition treatment; the adopted rule remains controlling. The timing difference is preserved as an implementation issue, and a final legal conclusion for a changed building requires the controlling authorities and its case record.

These route and cohort differences define the unit of comparison. Before administrative activity can be pooled or linked to a project, the analysis must identify the building’s applicable Article, cohort, proof type, and reporting period. Otherwise, a filing count can combine legally different events.

4.2 Accommodation is distributed across institutions

Within Article 320, reporting is accompanied by procedures for documentation, extensions, penalties, mitigation, and adjustments (New York City Department of Buildings, 2026b, § 103-14(b), (f)–(j)). An owner may apply for an adjustment on defined external-constraint or financial-constraint grounds. The maximum effective period is three calendar years for the former and one calendar year for the latter (New York City Department of Buildings, 2026a, § 103-12(b)(7), (c)(5)). The financial-constraint rule provides alternative showings. One branch requires an affidavit concerning work with a City-funded compliance resource and efforts to obtain applicable incentives and financing, together with specified offsets and financial metrics; another permits attestations concerning specified tax liens or HPD arrears (New York City Department of Buildings, 2026a, § 103-12(c)(3)–(4)). These provisions establish an application route, branch-specific documentary thresholds, and a possible period of relief.

Article 321 contains different mechanisms for extensions, penalty mitigation, eligible energy-conservation projects, and mediated resolution (New York City Department of Buildings, 2026c, § 103-17(d)–(g)). A mediated resolution is an agreement under which DOB does not bring an enforcement proceeding, potentially subject to terms such as a compliance plan; failure to meet those terms may lead to enforcement and civil penalties (New York City Department of Buildings, 2026c, § 103-17(g)). Here, the accommodation concerns enforcement, time, and required documentation within the Article 321 route.

Technical and financial support introduce additional institutions. HPD and NYC Accelerator describe technical assistance, compliance planning, service-provider navigation, and finance connections for specified building groups (New York City Department of Housing Preservation and Development, 2026; NYC Accelerator, 2026b,a). The AHRF rule covers qualifying electrification projects at buildings subject to affordable-housing regulatory agreements. It authorizes eligible offset purchases subject to a 10-percent limit on the building’s annual emissions limit and requires a registry for purchases, project assignment, retirement, and associated emissions reductions (New York City Department of Buildings, 2026b, § 103-14(e)(3)). These arrangements make assistance and financing routes available, but they are administered and documented outside the core annual emissions report.

Rent treatment is separate again. The current State rules for major capital improvements in rent-stabilized housing establish their own eligibility, application, order, notice, cap, duration, and audit framework (New York State Homes and Community Renewal, 2023, § 2522.4(b)); (New York State Homes and Community Renewal, 2025). Whether an owner sought or received a rent adjustment depends on a New York State Homes and Community Renewal record tied to the same building and project.

4.3 A bounded institutional trace

A stylized Article 320 pathway shows why this distribution is analytically consequential. Begin with a building’s dated CBL classification and the annual limit that applies to its route. If an owner invokes the financial-constraint adjustment, the applicable branch of the rule identifies the required showing; the application, requests for further information, and DOB disposition would show whether the route was used, how the agency decided, and how long the process took. Assistance or financing records could then show a referral, underwriting decision, or capital commitment. Contracts, invoices, installation records, and commissioning documents would establish project delivery. The applicable finalized annual record would establish verified emissions performance.

If the same project concerns rent-stabilized units and the owner seeks major-capital-improvement treatment, the relevant application and order arise in the separate State rent system. Owner financial records, rent and energy-cost measures, and resident evidence would then be needed to assess incidence. Each link answers a distinct question. The trace is stylized because the frozen corpus does not contain one public building file spanning every institution; its purpose is to identify the records that would make an actual case traceable.

The example exposes the substantive result. LL97’s affordability problem extends beyond the presence of a legal adjustment. It develops through the interaction of obligation, administrative time, access to finance, physical delivery, technical performance, rent treatment, and owner or resident conditions. Formal accommodation is distributed across those decision sites.

5 What the Frozen LL97 Corpus Shows

The proposition matrix makes the corpus’s evidentiary structure visible. Of 37 material LL97 proposition clusters, 25 concern formal design and 12 concern observable administrative activity; none directly establishes a stage 3–6 result. This documentary boundary marks the point at which analysis must move from public law, guidance, releases, and registries to linked case records.

5.1 Detailed design and visible early activity

Stage 1 is extensive because the law and rules specify obligations, proof, reporting, documentation, adjustment, mitigation, extensions, and mediated resolution in detail (City of New York, 2026a,b; New York City Department of Buildings, 2026a,b,c). Official materials also describe technical assistance and finance routes, while the AHRF materials define an offset mechanism

and intended registry (New York City Department of Housing Preservation and Development, 2026; NYC Accelerator, 2026b,a; New York City Department of Buildings, 2026d). Together, these sources make the front-end architecture legible across several institutions.

Stage 2 contains dated activity with heterogeneous denominators. In its April 22, 2026 release, DOB and HPD reported filings for approximately 93 percent of covered privately owned properties at the tax-lot level, representing 91 percent of covered buildings. The same release described filings from approximately 28,000 buildings as under audit and approximately 1,400 required properties as not having filed (New York City Department of Buildings and New York City Department of Housing Preservation and Development, 2026). The release placed these figures before completion of the audit process and stated that more data would follow. They measure filing and audit-stage activity for the populations defined in that release.

Other records illuminate narrower parts of implementation. The 2026 CBL assigns published compliance-path status and describes dispute and transition processes (New York City Department of Buildings, 2026e). DOB’s property inventory dictionary defines fields including Article 321 report receipt and selected ticket information (New York City Department of Buildings, 2025). AHRF rules and program materials define the registry’s intended fields, and the April 2026 registry snapshot records source-defined purchases, allocations, status labels, and reported information (New York City Department of Buildings, 2026b, § 103-14(e)(3)); (New York City Department of Buildings, 2026d; New York City Energy Efficiency Corporation, 2026).

Program contacts supply another activity measure. The City’s 2025 PlaNYC progress report stated that more than 1,000 owners began engaging with 321 Go between April 2024 and April 2025 and that nearly 400 received scope-review assistance (City of New York, Mayor’s Office of Climate and Environmental Justice, 2025, p. 12). Those counts describe different events from filing, audit, or registry activity. They remain analytically useful when their program definition, population, and period stay visible.

5.2 Where case linkage becomes necessary

Stage 3 begins when a formal route generates a case decision. An adjustment application, mediated resolution, approval or denial, request for information, and disposition date can show whether a building entered a process, what the agency decided, and how much time elapsed. Timing is part of the affordability analysis because an administrative interval may overlap with financing, procurement, or a later compliance milestone.

Article 321 illustrates the transition from decision to delivery. A mediated resolution may

include a compliance plan, and the eligible energy-conservation-project route may require a governmental-assistance commitment and supporting documents (New York City Department of Buildings, 2026c, § 103-17(f)–(g)). These records can establish a decision, planned scope, or documented commitment. Project delivery requires evidence of procurement, contract execution, installation, commissioning, payment, or physical completion tied to that case.

Verified performance presents a further route-specific question. An audited final determination and relevant annual-emissions result can establish Article 320 performance for the applicable period. Article 321 uses its own forms of proof. The April 2026 release documents filings under audit, and the CBL documents classification (New York City Department of Buildings and New York City Department of Housing Preservation and Development, 2026; New York City Department of Buildings, 2026e); neither is the final building-level performance result.

Owner and resident incidence then crosses into financial, rent, energy, service, health, participation, and housing records. Where rent-stabilized housing is involved, the State rent process supplies its own application and decision record (New York State Homes and Community Renewal, 2023, 2025). Linking that record to LL97 classification, administrative decisions, finance, and project delivery would create one analyzable building and intervention trace. Institutional adjacency alone leaves the connection untested.

The case therefore yields three findings. First, affordability-relevant accommodation is distributed across compliance, assistance, finance, delivery, and rent treatment. Second, Articles, cohorts, and classification dates produce non-comparable records unless route and period are preserved. Third, the transition from public design and activity records to performance and incidence is a longitudinal linkage problem. These substantive features of LL97’s governance have direct consequences for policy evaluation.

6 A Cohort-Sensitive Evaluation Design for 2030

The year 2030 matters because Article 320 begins a stricter emissions-limit period, while Article 321 remains a distinct energy-conservation-measure pathway. The 2026 CBL records pathway labels and transition treatment for specified status changes (City of New York, 2026a,b; New York City Department of Buildings, 2026e). The starting unit for evaluation is therefore a building-year or defined population with an applicable Article, cohort, reporting period, and any relevant reclassification.

The emissions–affordability question can then be divided into four linked parts. The first is capital and finance: what technical work is indicated, what it costs, how it is financed, and whether an adjustment or assistance route changes timing or scope. The second is delivery and

performance: what work proceeds, when it is completed, and what verified emissions result follows. The third is incidence: how financing, project operation, and any rent treatment affect owners and residents. The fourth is administrative timing and enforcement, which can interact with each of the other three. Table 2 translates those questions into observable records and outcomes.

Table 2: Operational design for evaluating the 2030 period

Analytical question	Identifiers and process records	Outcome evidence
Capital and finance	Building condition; Article, cohort, and applicable limit; project scope and cost; incentive, loan, assistance, or adjustment application and decision	Owner capital commitment, financing terms, approved scope, and dated financial position
Delivery and performance	Assessment, procurement, contract, installation, commissioning, and reporting milestones	Completed work, audited determination, and route- and period-specific emissions result
Owner and resident incidence	Housing and rent status; project and financing link; State rent application or order where relevant	Rent and energy costs, building service, comfort, health, participation, owner condition, and housing outcomes
Administrative timing and enforcement	Notice, response, review, request for information, mitigation or mediated resolution, adjudication, and disposition dates	Time to decision and owner-facing consequence for a defined obligation and case stage

Capital pressure depends on building-specific conditions. If a building facing a stricter applicable limit cannot meet it through operations alone, the owner may have to choose among technical scope, sequencing, and financing options. The magnitude and distribution of that pressure depend on building condition, ownership and contract structure, borrowing terms, incentives, and timing. Scenario research shows that target timing and scope can change modeled emissions pathways, while penalty studies estimate financial exposure only under stated local assumptions (Walter and Mathew, 2021; Taylor et al., 2025). A New York analysis must therefore use building-specific technical and financial inputs.

Delivery requires a dated project sequence from assessment through commissioning and verified performance. Retrofit research shows that program design and population characteristics can shape subsidy participation and recorded program outputs (Willand et al., 2020). Data systems designed for reporting may omit the contextual fields needed to observe that sequence (Syse et al., 2026). Linking project milestones to the applicable LL97 route makes it possible to distinguish an administrative delay, a financing delay, a change in scope, completed work, and a technical result.

Incidence requires another join. In rental housing, the parties who pay for capital work and experience changes in operating cost or comfort may differ. The direction and magnitude

of rental-market effects are empirical questions that require jurisdiction-specific, linked data (Singhal et al., 2025; Fuerst et al., 2026). Under LL97, an incidence analysis would connect the building’s route and project to financing, the State rent process where relevant, energy expenditures, service conditions, and direct owner or resident measures over time.

The current rules identify where several of these paths begin. They describe reporting, adjustment, extensions, mitigation, and mediated resolution (New York City Department of Buildings, 2026a,b,c). HPD and NYC Accelerator separately describe technical-assistance and finance routes (New York City Department of Housing Preservation and Development, 2026; NYC Accelerator, 2026b,a). The April 2026 release supplies an early observation point through its filing and audit-stage measures (New York City Department of Buildings and New York City Department of Housing Preservation and Development, 2026). Following a cohort from that point would require stable case identifiers and dates across DOB, assistance and finance providers, project records, verified performance, and any State rent record.

A descriptive implementation study can use those linked sequences to measure participation, decisions, timing, delivery, and performance. Estimating effects would require an additional temporal or comparison design suited to the outcome. Distributional analysis must also define whose outcome is measured and how it represents the relevant energy-justice dimension (Baker et al., 2023). The framework identifies the compatible records from which those designs can be built.

The frozen corpus supports a concrete evaluation design: identify the cohort and route, follow decisions and projects over time, verify route-specific performance, and connect owner and resident outcomes through matched institutional records. A directional 2030 verdict depends on the linked outcome evidence generated by that design.

7 Discussion and Conclusion

LL97 requires a linked institutional sequence for joint evaluation of emissions and affordability. The emissions obligation, compliance route, assistance, financing, project delivery, rent treatment, and owner or resident experience are administered in different places and recorded at different stages. A joint assessment must follow the same building or defined population, legal route, and period across those systems.

Three findings support that conclusion. First, affordability-relevant accommodation is distributed across DOB compliance and adjustment procedures, City assistance and finance, project delivery, and a legally separate State rent-treatment process. Second, Article, cohort, and classification differences create distinct obligations and forms of proof. Third, the frozen corpus is rich in formal design and visible early activity, while later questions become case-

linkage problems. The proposition matrix makes the last point concrete: 25 clusters concern formal design and 12 concern administrative activity. These counts establish the documentary boundary between the present analysis and an outcome study of emissions or affordability.

The framework’s contribution lies in separating evidence stage from four other tests: authority and provenance, legal route and cohort, temporal fit, and linkage. This resolves an ambiguity that a simple ladder of outputs and outcomes would leave in place. A rule may be the best source for a legal proposition while remaining irrelevant to whether a project was delivered. A resident account may directly observe lived experience while saying nothing about the controlling compliance obligation. Both can enter a joint analysis when the case, route, and period connect them.

For LL97, that distinction keeps the housing problem substantive. An adjustment provision is part of affordability-relevant design, but affordability also depends on administrative time, access to capital, completed work, technical performance, the separate treatment of rent, and changes experienced by owners and residents. Accommodation is design evidence; resolving burdens or identifying hardship requires matched outcome evidence. The framework specifies how those questions can be answered.

The 2030 design follows directly from the case. Researchers can begin with route- and cohort-specific building populations, preserve filing and decision dates, and link project and performance records before adding financial, rent, energy-cost, or resident measures. Where causal effects are the question, those linked data must be paired with an appropriate comparison or temporal strategy. Where justice is the question, the outcome and affected population must correspond to the distributive, procedural, or recognition dimension under study (Jenkins et al., 2016; Baker et al., 2023).

Public reporting can support that work when it preserves stable identifiers, proof type, denominator, and time. The April 2026 release is useful because it documents a defined filing and audit stage (New York City Department of Buildings and New York City Department of Housing Preservation and Development, 2026). Later releases and case records would become more analytically valuable if they could be connected to decisions, projects, verified performance, and relevant owner or resident outcomes. This evaluative capability allows climate and affordability questions to be studied together across the full implementation process.

The study is limited to one city, a proposition-selected public corpus, and a documentary design; confidential building files, interviews, direct resident experience, and causal effects fall outside its evidence base. Independent application would be needed to establish how well the compatibility tests travel to other policies. Their conditional relevance is greatest where an environmental mandate also divides obligations, accommodations, delivery, and outcomes

across legal cohorts and institutions.

As LL97 moves toward the stricter 2030 period, the immediate contribution is an observation design. It identifies what the current case evidence establishes, what records can be compared, and how future research can assess emissions performance and affordability within the same implementation trace. A directional verdict depends on the linked outcome evidence that design is meant to organize.

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